

Forecasting the Macroeconomy

WEEK 3

FINS5545: FINANCIAL MARKET DATA LITERACY

LEARNING OBJECTIVES (1/2)

By the end of this lecture, you will be able to:

- Describe the three **rules** for forecasting in this course — the modelling target is the *monthly change*, evaluation is in *target space*, the split is in-sample through *2019-12-31* and out-of-sample from *2020-01-31*
- Test whether a macro time series should be modelled in levels or first differences using the Augmented Dickey–Fuller (ADF) test, calibrated against simulated random-walk and stationary AR(1) series whose true behaviour is known to us in advance
- Locate forecasting on the Data Factory Floor (DFF) — Station 3 (Model Design)

LEARNING OBJECTIVES (2/2)

You will also be able to:

- Implement the Australian sequence of forecasting models in Python — naive, $AR(p)$, $ARMA(p,q)$, ARX, ARMAX, OLS dynamic regression, and elastic net — with lagged exogenous predictors that respect the release-lag rule
- Evaluate competing forecasts with target-space MAE, RMSE, MASE, and out-of-sample R^2 versus the naive benchmark; rank models on a horse-race leaderboard
- Build Case Study 1 on Australia and mirror the workflow on Case Study 2 on the United States
- Apply the workflow to a macro variable of your own choosing

REQUIRED READING

Course-mandated textbook

Hric and Lin (2026). *Applied Data Science in FinTech: Models, Tools, and Case Studies*, **Chapter 4** (“Data Modeling in FinTech”).

These slides and the in-house chapter are the *companion* material. The lecture works the same forecasting workflow on the Australian unemployment rate, then mirrors it on the United States, then hands the workflow to you.

Background references (not mandated)

- Hyndman and Athanasopoulos (2021), *Forecasting: Principles and Practice*, 3rd ed. — free online at <https://otexts.com/fpp3/>
- Diebold (2024), *Forecasting in Economics, Business, Finance and Beyond* — free online (UPenn)
- Stock and Watson (2020), *Introduction to Econometrics*, 4th ed. — standard reference for ADL/ARX models and BIC lag selection
- Hastie et al. (2009), *The Elements of Statistical Learning*, 2nd ed., §3.4 — elastic net and the bias-variance tradeoff

AUDIENCES FOR MACRO FORECASTS

WHO USES MACRO FORECASTS?

Six audiences depend on the same kind of forecast that this week trains you to build:

- **Central banks** (RBA, Federal Reserve, ECB, Bank of England) set policy interest rates based on forecasts of inflation and unemployment. The RBA publishes its central forecasts every quarter in the *Statement on Monetary Policy* (Reserve Bank of Australia, 2026).
- **Treasuries and finance ministries** build budgets on forecasts of GDP growth and inflation.
- **Bond markets** price the yield curve on expected paths for inflation and the policy rate.
- **Banks** size loan books and price credit on unemployment and GDP growth forecasts.
- **Corporations** make hiring, capital-expenditure, and inventory decisions on the same macro forecasts.
- **Macro hedge funds and asset allocators** take cross-country positions based on these forecasts.

WHAT IS AT STAKE

A wrong macroeconomic forecast leads to a wrong policy decision, and the cost falls on the real economy.

- **Reserve Bank of Australia, 2021.** The RBA communicated that the cash rate would stay near zero “until 2024”. Inflation surprised on the upside; the cash rate rose from 0.10% in May 2022 to over 4% by late 2023 — about **4 percentage points** in 18 months. Mortgage repayments rose sharply as a direct consequence.
- **Consequences for the RBA.** Governor Philip Lowe was *not reappointed* when his term ended in September 2023; the forward-guidance miscommunication was widely cited as the reason. Michele Bullock became the first woman to hold the position. A separately commissioned RBA Review then reformed how the Bank produces and communicates its forecasts.
- **Bank of England, 2024.** The Bank commissioned a public review of its forecasting framework (Bernanke, 2024) after a similar string of inflation misses. The review’s headline recommendation was to retire single-point “fan chart” projections in favour of named alternative *scenarios*.

The same forecasting workflow runs at central banks, banks, funds, and corporations, and getting it right determines real decisions, careers, and institutional reforms.

FOUR PARTS OF A COMPLETE FORECAST

A single number is an incomplete forecast. A complete forecast contains four parts:

1. A **central case** — the point forecast.
2. A **measure of the uncertainty** around the central case (target-space metrics, prediction intervals).
3. **Sensitivity to assumptions** — what changes if a key assumption moves. These are the *scenarios* Bernanke (2024) recommended for the Bank of England.
4. An **update path** — how the forecast is revised when new information arrives.

A forecaster is not judged on whether their central case was correct, but on whether their forecast led to decisions that remained reasonable even when the forecast turned out to be wrong. Elliott and Timmermann (2008) frame this directly: the user's *loss function* determines what makes a forecast useful.

U.S. FORECASTING BENCHMARKS

Real-world forecasts are public, comparable, and tracked over time. Four standard U.S.-market reference points (Australia's nearest analogues are the RBA *Statement on Monetary Policy* and the ABS data-release programme):

- **Survey of Professional Forecasters (SPF)** (Federal Reserve Bank of Philadelphia, 2026). Quarterly survey of about 40 economists since 1968 (Philadelphia Fed took it over from ASA/NBER in 1990). Publishes the mean, the median, and an *anxious index* (probability of a real-GDP decline).
- **Federal Reserve Tealbook** (formerly Greenbook). Internal forecasts produced before every FOMC meeting; released publicly after a five-year lag.
- **Atlanta Fed GDPNow** (Federal Reserve Bank of Atlanta, 2026; Higgins, 2014). A real-time **nowcast** of U.S. quarterly GDP growth, updated as new monthly indicators arrive.
- **FRED-MD and FRED-QD** (McCracken and Ng, 2016; McCracken and Ng, 2021). Public macro databases that mirror the data central-bank research staff use; foundational for academic forecasting research.

Your assignment is judged on the same logic: a public target, a public benchmark, and a track record.

FORECASTING ON THE DATA FACTORY FLOOR

THE DATA FACTORY FLOOR (DFF)

The DFF map introduced in Week 1 has four stations:

- **Station 1** (Data Lake / ETL) — download raw data, type it, audit it
- **Station 2** (Feature Engineering) — transform series into the right units for the question, build a tidy panel
- **Station 3** (Model Design) — choose a target, fit candidate models, evaluate out of sample *this is Week 3*
- **Station 4** (Implementation) — deploy the chosen model, monitor it, retrain when needed

Stations 1 and 2 are already done by the time you arrive at Station 3. The Australia case uses the Stage 1 long table from Week 2 directly. The U.S. case follows the same template on a Federal Reserve Economic Data (FRED) panel.

STATION 3 IN THE REQUIRED READING

Chapter 4 of Hric and Lin (2026) walks through Station 3 with a different example: forecasting household energy use for a smart-home (IoT) product. Their workflow is the same as ours:

1. Receive a prepared data frame from Station 2.
2. Choose a target (kilowatt-hours per day, in their case).
3. Fit candidate models, including an $\text{ARIMA}(p, d, q)$ specification.
4. Score model accuracy on held-out data.
5. Hand the chosen model to Station 4 for use in the product.

The ARIMA notation in the book maps directly to the models we use this week: the AR component is our $\text{AR}(p)$, the I (integration) component is the differencing we already did to get the change target, and the MA component is the MA part of our ARMA.

STATION 3 IN TWO DOMAINS

Two Station 3 walk-throughs, side by side:

- **Hric & Lin, Chapter 4.** Domain: smart-home Internet-of-Things product. Target: daily energy use in kilowatt-hours. Model family: $\text{ARIMA}(p, d, q)$ plus an LSTM neural-network extension. Deliverable: a forecast feed for the product dashboard.
- **This week.** Domain: macroeconomic forecasting for monetary policy and asset allocation. Target: monthly change in the unemployment rate. Model family: naive, AR, ARMA, ARX, ARMAX, OLS, elastic net. Deliverable: a target-space leaderboard with five FT-style narrative figures.

The DFF Station 3 logic — target → benchmark → candidate models → out-of-sample evaluation → hand-off — is identical. The domain, the data frequency, and the chosen model family change, but the workflow does not.

THE THREE RULES OF THIS CHAPTER

Three rules that hold throughout this chapter:

1. **Target = monthly change.** We forecast the change in the unemployment rate, not the level. Levels of macro series usually drift; changes are far closer to stationary.
2. **Evaluation is in target space.** We score every model on its forecast of the change — target MAE, RMSE, MASE, out-of-sample R^2 versus the naive benchmark. The level path is for communication only.
3. **Default split.** Train in-sample through 2019-12-31; test out-of-sample from 2020-01-31 onward. Same split for the Australia case, the United States case, and the assignment baseline.

If you cannot say what your target is, how you are evaluating, and what your benchmark is, you do not yet have a forecast.

PLAN FOR THE WEEK

Four deliverables, in this order:

- **Australia case study:** forecast the monthly change in Australia's unemployment rate, one step ahead. Full sequence of models: naive, AR, ARMA, ARX, ARMAX, OLS, elastic net, horse race, ensemble.
- **United States case study:** forecast the monthly change in the U.S. unemployment rate, one step ahead. Shorter sequence: naive, AR, ARMA, ARX, horse race. The same workflow on a second market.
- **Assignment (your choice):** pick a non-unemployment macro variable from the Australian or U.S. data we provide and repeat the workflow on your chosen target.
- **Deploy the apps:** publish both Week 3 Streamlit apps to Streamlit Community Cloud once the lecture content is complete.

WHAT IS A FORECAST?

THE FORECASTING TASK

A **forecast** is a statement about a future value of a variable, made using the information available today. In this chapter every forecast is:

- a **point forecast** (one number, not a probability distribution)
- at **horizon** $h = 1$ (one step ahead: the next month for our monthly target)
- built from information available at time t only (no use of data dated after t)

Notation: $\hat{y}_{t+1|t}$ is our forecast for y_{t+1} made at time t . The error is $e_{t+1} = y_{t+1} - \hat{y}_{t+1|t}$. We average those errors to score the model.

FORECASTING VS EXPLAINING

In an econometrics class, a regression often *explains* a relationship: we estimate β and interpret its sign and magnitude. The fit is judged *in sample*.

A **forecast** is judged differently. The relevant question is not “does the model fit the past?” but “does the model predict data it has not seen?”

Diebold (2024) is direct about this: a forecasting model is judged on data it has not seen, not on how well it fits the data it was trained on. A model that fits 80% of the in-sample variance but cannot beat a no-change rule out of sample is not useful for forecasting.

In-sample R^2 is not a forecast metric.

WHAT INFORMATION IS AVAILABLE AT TIME t

When the forecast for y_{t+1} is built, the only information we are allowed to use is information *observable at time t* . In macro forecasting this is stricter than it sounds:

- Australian gross domestic product (GDP) for the September quarter is released in early December. If your forecast is dated 2024-11-30, you may not use it.
- Headline CPI for a given quarter is released about one month later. The Stage 1 long table from Week 2 records this with the `observable_month_end` field.
- Every exogenous predictor in this week's models enters at $t - 1$, not at t . A same-month predictor is not yet observable when the forecast is made, and using it inflates every metric.

WHY START WITH THE NAIVE BENCHMARK

The first model in every forecasting exercise is the **naive benchmark**. For a change target it is:

$$\hat{y}_{t+1|t} = 0$$

Forecast: “no change”. It is a useful baseline for three reasons.

- It is the simplest model that respects the change target. If a richer model cannot beat it, the added complexity is not earned.
- It is the reference scale for the unit-free metrics (MASE, out-of-sample R^2) used to compare models.
- It tells the audience how much a richer model improves on the simplest possible baseline.

TARGET AND TRANSFORMATION

AUSTRALIAN UNEMPLOYMENT RATE: THE LEVEL

Our running example is the Australian unemployment rate, monthly, seasonally adjusted, from the Reserve Bank of Australia (RBA) Stage 1 fixture. The data goes back to the early 1980s and through December 2025.

Plotted as a level, the series:

- drifts up and down for long periods (1990s recession, 2000s mining boom, COVID-19 shock, post-COVID tightening)
- is visibly persistent — this month is close to last month
- shows no obvious mean to revert to

This is the classic shape of a series that should not be modelled in levels. We need a transformation that converts it into something close to stationary.

AUSTRALIAN UNEMPLOYMENT RATE: THE MONTHLY CHANGE

Take first differences:

$$\Delta y_t = y_t - y_{t-1}$$

This is the **monthly change** in the unemployment rate, in percentage points (pp). It is the actual target of every model in this chapter.

Plotted, the change series:

- moves around zero (small positives in slow months, occasional sharp positives during recessions)
- has no obvious trend
- shows much weaker persistence than the level

WHY A UNIT-ROOT LEVEL IS DIFFICULT TO FORECAST

- A unit-root series has **no mean it must return to**. The Australian unemployment rate today is the sum of *every monthly shock* the labour market has received since the series began.
- To forecast tomorrow's *level*, we would need to forecast tomorrow's *shock* on top of today's level. By definition, shocks are unpredictable. The best forecast of the level is therefore *today's level* — the no-change rule.
- The *change* has structure we can study: a typical size of about 0.1 pp per month, a stable mean near zero, and modest persistence around recessions and recoveries. There is signal here a model can capture.
- Regressions fit on two unit-root levels are misleading. Two unrelated random walks can show $R^2 \approx 0.9$ by accident — this is **spurious regression**. Forecasting the change avoids this trap.

Forecasting the change is forecasting the predictable part of the series.

THE AUGMENTED DICKEY–FULLER TEST

The **Augmented Dickey–Fuller (ADF)** test, due to Dickey and Fuller (1979), tests for a unit root — the case where a level series has $\rho = 1$ in

$$y_t = \rho y_{t-1} + \varepsilon_t$$

Under $\rho = 1$ the series is a random walk: it drifts forever and is not stationary.

Practical reading: a **low p -value** (below 0.05) is evidence *against* a unit root — the series looks stationary. A **high p -value** means we cannot reject the unit-root hypothesis; the series probably needs differencing.

Apply the ADF test to both the level and the first difference. Pick the form where the test passes.

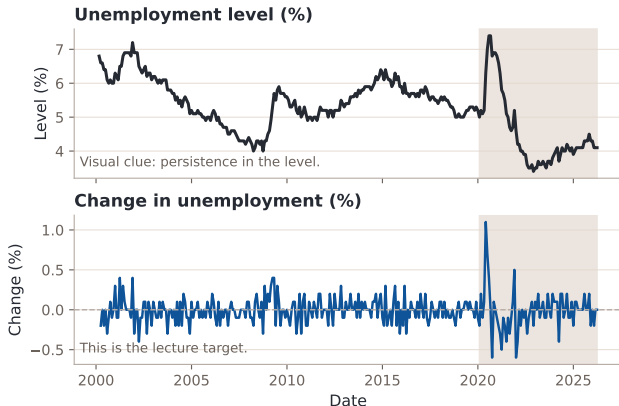
ADF ON SIMULATED SERIES

Before we trust the ADF on real data, we calibrate it on two simulated series whose data-generating process is known to us:

- **Random walk:** $y_t = y_{t-1} + \varepsilon_t$. The level should fail the ADF test. The first difference should pass.
- **Stationary AR(1) around zero:** $y_t = 0.6 y_{t-1} + \varepsilon_t$. The level should pass.

Why bother? The ADF test is not magic — it has known weaknesses for borderline cases. Simulating two series whose answer we already know is a quick one-line check before we run it on real data. If the ADF test gives the right answer on the simulated series, we trust it more on the real one.

ADF ON AUSTRALIAN UNEMPLOYMENT



Out-of-sample evaluation window shaded | Sample: 2000-02-29 to 2026-03-31 | Week 3 fixture data

Monthly, seasonally adjusted; RBA H5.

- Top: *level* drifts up in the 1990s, falls in the 2000s, jumps in 2020, hits historic lows in 2022.
- No mean it must return to — the unit-root signature.
- Bottom: *first difference* moves around zero with stable variance.
- ADF on the level: high p , cannot reject unit root.
- ADF on the change: low p , reject unit root.
- Verdict: model the change.

THE NAIVE BENCHMARK AND TARGET-SPACE METRICS

THE NAIVE FORECAST

In target space the naive forecast is one line of code:

$$\widehat{\Delta y}_{t+1|t} = 0$$

Equivalently: next month's unemployment rate is forecast to equal this month's. Simple to compute, free to deploy, and the benchmark every richer model must beat.

It is also a strong baseline. In macro forecasting, the level of unemployment, inflation, output, or interest rates tomorrow is often closer to today than to anything a complicated model can produce. That is the reason we always include it.

ONE-STEP ROLLING EVALUATION

We do not score a model on a single forecast. We score it on a **rolling window** of one-step-ahead forecasts:

- Fit the model on data through month t .
- Forecast month $t + 1$. Record the error.
- Move forward: refit through $t + 1$, forecast $t + 2$, record. Repeat.

Over the test window we accumulate one error per month. Those errors are what target-space MAE, RMSE, MASE, and out-of-sample R^2 are computed from.

THE TRAIN/TEST SPLIT

Every model in this chapter uses the same split:

- **Train (in-sample):** data up to and including 2019-12-31.
- **Test (out-of-sample):** data from 2020-01-31 onward.

The split is deliberate. The test window spans COVID-19, the post-COVID inflation surge, and the tightening cycle that followed. That is a real macro stress test — exactly the period where a forecaster's model is most likely to be wrong.

When you build your own forecast in the assignment, you may extend the test window forward, but you may not move the training endpoint past 2019-12-31 without an explicit reason.

FOUR TARGET-SPACE METRICS

Four numbers per model, all computed on the change target Δy :

- **MAE** — mean absolute error. Average $|e_{t+1}|$ across the test window. Same units as the target (pp).
- **RMSE** — root mean squared error. Punishes big errors more than MAE. Used to rank models.
- **MASE** — mean absolute scaled error (Hyndman and Koehler, 2006). MAE scaled by the in-sample naive MAE. Unit-free. Below 1 means “better than naive”.
- **Out-of-sample R^2 vs naive**. $1 - \text{RMSE}^2 / \text{RMSE}_{\text{naive}}^2$. Above 0 means “beats naive”. The headline metric for a finance audience.

Each metric tells you something different:

- **MAE** is the typical-sized miss. “On average we are off by X pp.”
- **RMSE** is the right metric when big misses hurt more than small ones (most macro applications).
- **MASE** is the right metric when comparing across very different series, because it is unit-free.
- **Out-of-sample R^2 vs naive** is the right metric for telling a non-technical reader whether the model has improved on doing nothing. “We explain 30% of the out-of-sample variance that naive does not.”

THE NAIVE BENCHMARK RULE

If a model does not beat the naive benchmark on the target,
out of sample, on this split,
do not use the model.

AR AND ARMA

The first richer model is an autoregression of order 1:

$$\Delta y_t = \alpha + \phi_1 \Delta y_{t-1} + \varepsilon_t$$

Why AR(1)? If the change in unemployment has any month-to-month persistence at all — if a positive change tends to be followed by another positive change — then AR(1) beats naive.

The naive forecast is the special case $\alpha = \phi_1 = 0$. AR(1) is the smallest deviation from naive a forecaster can make.

CHOOSING p WITH BIC

More lags can be added:

$$\Delta y_t = \alpha + \phi_1 \Delta y_{t-1} + \phi_2 \Delta y_{t-2} + \cdots + \phi_p \Delta y_{t-p} + \varepsilon_t$$

But how many? The **Bayesian information criterion (BIC)** picks the lag order that trades fit against complexity:

$$\text{BIC} = -2 \ln L + p \ln T$$

An extra lag is added only if it improves the log-likelihood enough to offset the $\ln T$ penalty. Stock and Watson (2020) discuss this trade-off in detail. The lecture grid is $p \in \{1, 3, 6, 12\}$.

AR(p) IN PYTHON

```
from statsmodels.tsa.ar_model import AutoReg

best_bic, best_p, best_model = float("inf"), None, None
for p in [1, 3, 6, 12]:
    model = AutoReg(train_change, lags=p, old_names=False).fit()
    if model.bic < best_bic:
        best_bic, best_p, best_model = model.bic, p, model

print(f"Selected lag order: {best_p}")
```

- Candidate grid is small and named — defensible in front of an audience.
- BIC penalty $p \ln T$ rules out wide lag orders on this sample length.
- Model is refit each month as the rolling window advances; the lag order is reselected each time.
- Reselecting prevents the lag order from drifting silently through the sample.

ARMA(p, q)

Autoregressive terms capture persistence in the level. Moving-average terms capture persistence in the *shocks*:

$$\Delta y_t = \alpha + \sum_{i=1}^p \phi_i \Delta y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

The Box–Jenkins workflow (Box et al., 2015) pairs AR and MA terms to capture both kinds of persistence with fewer parameters than either alone.

ARMA ORDER SELECTION

We do not search every (p, q) . The lecture uses a small, named grid:

- $(1, 0)$ — AR(1), no MA
- $(2, 0)$ — AR(2), no MA
- $(1, 1)$ — one AR term, one MA term
- $(2, 1)$ — two AR terms, one MA term

BIC selects within the grid. Keeping the grid small makes the model straightforward to defend in front of an audience: “we picked from four candidates, BIC chose this one, and here is why”.

After the winning ARMA is fit, check its residuals:

- Plot $\hat{\varepsilon}_t$ over time. Should look like noise — no trend, no widening or narrowing variance.
- Plot the autocorrelation (ACF) of $\hat{\varepsilon}_t$. Should be within the confidence bands at every lag.
- Plot the partial autocorrelation (PACF). Same standard.

A pattern in the residuals is a sign the model has missed predictable content. In the lecture code, the diagnostic plots are produced automatically; in your assignment they are part of the deliverable.

ARX AND ARMAX

ARX: ADDING ONE OUTSIDE VARIABLE

An autoregression on its own can only learn from its own past. To use information about the rest of the economy, we add an **exogenous** predictor:

$$\Delta y_t = \alpha + \phi_1 \Delta y_{t-1} + \beta_1 x_{t-1} + \varepsilon_t$$

For the Australia case, the first outside variable is `cash_rate_change_pp` — last month's change in the RBA cash rate, in percentage points.

The economic intuition: monetary policy moves first; the labour market responds with a lag. A 25-basis-point cash-rate change in March may show up in unemployment in April or May.

Every exogenous predictor enters at $t - 1$, not t . Three reasons.

- **No look-ahead.** At the forecast date t we know the cash rate at $t - 1$. We do not yet know it at t until it is released.
- **Valid prediction.** Using x_t to forecast y_{t+1} at time t may be valid, but using x_t to forecast y_t is contemporaneous, not predictive — it explains, it does not forecast.
- **Defensibility.** A real-time forecaster has to honour publication lags. The `observable_month_end` field from Week 2 makes the lag bookkeeping explicit.

ARMAX: ADDING A SECOND OUTSIDE VARIABLE

The ARMA backbone takes the same exogenous predictors:

$$\Delta y_t = \alpha + \sum_{i=1}^p \phi_i \Delta y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \boldsymbol{\beta}^\top \mathbf{x}_{t-1} + \varepsilon_t$$

The lecture ARMAX has $(p, q) = (1, 0)$ and *two* lagged outside variables:

- `cash_rate_change_pp` — monetary policy stance
- `commodity_price_log_change_pct` — last month's log change in the A\$ commodity price index, an external-demand signal

WHY PREDICTORS MUST BE LAGGED

A common mistake in student code:

```
X = panel[["cash_rate_change_pp"]] # WRONG -- this is  $x_t$ , not  $x_{t-1}$ 
```

Fitting this puts contemporaneous information into the predictor. The model will appear strong in sample. Out of sample, when the predictor cannot be observed at the forecast date, the metrics collapse.

Rule: build the design matrix by explicit `.shift(1)` of every exogenous column. Verify by checking that the maximum date of X is one month earlier than the maximum date of y .

OLS REGRESSION

ARX AS A REGRESSION

An ARX model is a regression of Δy_t on its own lags and lagged exogenous variables. So is an ordinary least squares (OLS) regression with the same predictors. The difference is presentation, not substance.

Why bother reframing?

- The OLS view scales to many predictors more naturally than the time-series view.
- The OLS view connects to the rest of the econometrics curriculum: $\hat{\beta}$, R^2 , standard errors, confidence intervals.
- The OLS view sets up the next step: shrinkage methods like the elastic net are just OLS with a penalty added.

THE OLS PREDICTOR SET

The lecture OLS uses target lags and five named macro predictors:

- Target lags: Δy_{t-1} , Δy_{t-3} , Δy_{t-6} , Δy_{t-12}
- Exogenous predictors (each at $t - 1$):
 - ▶ `cash_rate_change_pp`
 - ▶ `commodity_price_log_change_pct`
 - ▶ `trade_weighted_index_log_change_pct`
 - ▶ `participation_change_pp`
 - ▶ `vacancies_to_labour_force_change_pp`

Nine predictors in total. Small enough to interpret, and large enough to give OLS a meaningful chance of beating the naive benchmark.

OLS IN PYTHON

```
import pandas as pd, statsmodels.api as sm

lag_cols = {f"lag_{k}": panel["unemployment_change_pp"].shift(k)
            for k in [1, 3, 6, 12]}
exog_cols = {
    name: panel[name].shift(1)  # the lag discipline
    for name in [
        "cash_rate_change_pp", "commodity_price_log_change_pct",
        "trade_weighted_index_log_change_pct",
        "participation_change_pp", "vacancies_to_labour_force_change_pp"
    ]
}
X = pd.DataFrame(**lag_cols, **exog_cols).dropna()
y = panel["unemployment_change_pp"].loc[X.index]
model = sm.OLS(y, sm.add_constant(X)).fit()
```

- Four target lags plus five lagged exogenous predictors = nine columns.
- Every exogenous column `.shift(1)` — no same-month values.
- `dropna()` drops warm-up rows; `y` aligned to `X`; refit each month.

IN-SAMPLE VS OUT-OF-SAMPLE R^2

Fit the OLS on the training window through 2019-12-31. Two numbers come out:

- In-sample R^2 : typically around **30–40%** for this specification.
- Out-of-sample R^2 vs naive on the COVID-and-after window: substantially lower (or even negative for some students).

The wedge between the two numbers is the cost of fitting. The more predictors you add, the bigger the wedge tends to get — the model fits the in-sample noise alongside the in-sample signal. This is the problem the elastic net is designed to solve.

SHRINKAGE AND THE ELASTIC NET

WHY OLS OVERFITS WITH MANY PREDICTORS

Add more macro predictors to the OLS regression:

- `employment_to_population_change_pp`
- `headline_cpi_inflation`, `trimmed_mean_inflation`, `wage_price_index_growth`
- U.S. spillover block: `us_fedfunds_change_pp`, `us_unemployment_change_pp`,
`us_indpro_log_growth_pct`, `us_vix_level`, `us_yield_spread_pp`

Now there are 14–15 predictors. OLS will estimate a coefficient for each one. In sample, R^2 rises mechanically. Out of sample, the metrics often degrade — the model has fit noise.

BIAS AND VARIANCE

Every estimator has two sources of error:

- **Bias** — on average, how far is the estimator from the truth?
- **Variance** — how much does the estimate move from sample to sample?

OLS is unbiased but **high variance** when predictors are correlated or the sample is short. A model with a small bit of bias and a much smaller variance can predict better out of sample.

Hastie et al. (2009) §2.9 frames this directly: $\text{prediction error} = \text{bias}^2 + \text{variance} + \text{irreducible noise}$. Shrinkage trades a small bias for a large drop in variance.

The **elastic net** (Zou and Hastie, 2005) replaces the OLS loss with:

$$\min_{\beta} \frac{1}{2N} \|y - X\beta\|_2^2 + \alpha \left(\rho \|\beta\|_1 + \frac{1-\rho}{2} \|\beta\|_2^2 \right)$$

Two effects:

- The L_1 term ($\|\beta\|_1$) shrinks coefficients toward zero and zeros out the weakest ones — this is the lasso effect.
- The L_2 term ($\|\beta\|_2^2$) handles correlated predictors gracefully — groups of correlated regressors are kept or dropped together.

Same predictor bank as OLS-plus-extras; smaller coefficients; lower variance; usually better out-of-sample.

CROSS-VALIDATION FOR α

The penalty strength α is not estimated — it is chosen. The correct way to choose it is **time-series cross-validation**:

- Hold out the last 12 months of training data.
- For each candidate α in $\{0.001, 0.01, 0.1, 1.0\}$, fit the elastic net on the remaining training data, score the held-out 12 months.
- Pick the α with the best out-of-sample R^2 on the validation window.
- Refit on the full training window with the chosen α .

THE HORSE RACE

ALL MODELS ON THE SAME HOLD-OUT WINDOW

Now we have the full set of forecasting models:

- Naive (change = 0)
- AR(1), AR(p^*) with BIC
- ARMA selected from $\{(1, 0), (2, 0), (1, 1), (2, 1)\}$
- ARX(1) with cash-rate change at $t - 1$
- ARMAX(1,0) with cash + commodity at $t - 1$
- OLS dynamic regression (9 predictors)
- Elastic net (14 predictors, CV-selected α)

Each is fit on the training window, evaluated on the same out-of-sample window starting 2020-01-31, and scored on the same four target-space metrics. The result is a **leaderboard**.

THE AUSTRALIAN LEADERBOARD

Model evaluation scorecard



Lower is better for RMSE, MAE, and MASE. Higher OOS R^2 means a larger forecast-error reduction versus naive.

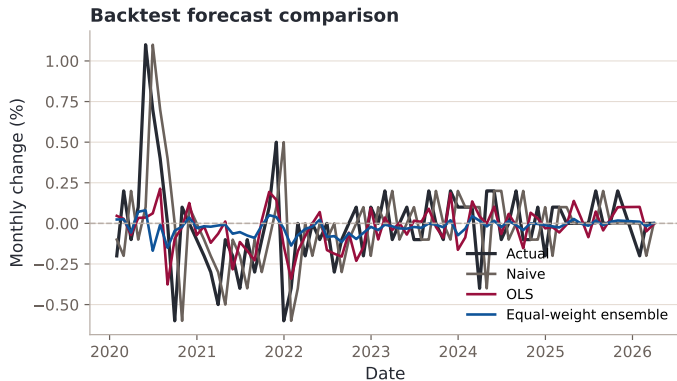
Eight models, same hold-out window, ranked by target RMSE.

- **OLS wins:** target RMSE ≈ 0.229 pp; OOS R^2 vs naive ≈ 0.43 .
- Elastic net is a very close second.
- ARMAX, ARX, ARMA, AR(1) trail OLS but still beat naive.
- Naive sits at the bottom under this split.
- Every richer model has OOS $R^2 > 0$.
- Source:
run_beginner_model_horse_race.py
on the Dec 2025 fixture.

READING THE LEADERBOARD

Three questions to ask of any leaderboard:

- **Which models beat the naive baseline?** Everything with target $\text{OOS-}R^2$ above zero. Under this split, every richer model in the AU pack does.
- **Is the gap economically meaningful?** OLS posts target $\text{OOS-}R^2 \approx 0.43$. The model explains 43% of the variance the naive forecaster does not. That is large by macro standards.
- **Does adding complexity improve performance enough to justify it?** The elastic net posts a very similar score with more predictors; ARMAX and ARX trail OLS. The lesson: complexity must be justified by an out-of-sample gain every time.

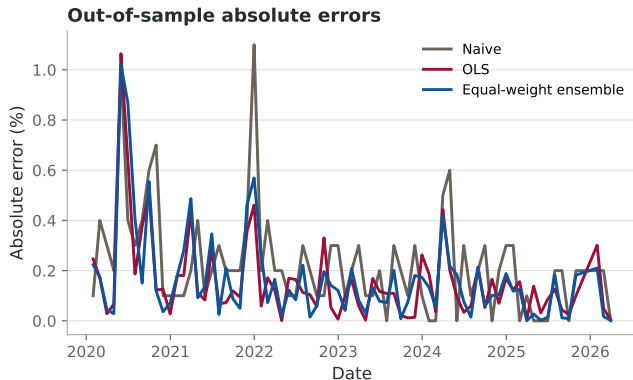


Out-of-sample only | Common evaluation dates: 73 months

Forecast paths vs realised change,
Jan 2020 – Dec 2025.

- OLS path tracks the realised series most closely.
- Naive is the flat line at zero — visibly off during the COVID shock.
- Misses concentrate at the COVID shock and the 2022 inflation surge.
- Ranking is otherwise stable across the test window.
- One forecast per month from the rolling one-step-ahead evaluation.

ABSOLUTE ERRORS



Lower lines mean smaller misses in the out-of-sample evaluation window

$|e_{t+1}|$ over the test window, model by model.

- April 2020 is the largest miss for every model.
- OLS has the lowest typical miss outside the COVID spike.
- Naive shows the widest band, as expected.
- Ranking by median $|e|$ matches the RMSE ranking.
- No model recovers the full COVID jump — April 2020 is a hard miss for any one-step-ahead forecaster.

FORECAST COMBINATION

THE EQUAL-WEIGHT AVERAGE

After fitting and scoring the individual models, average their forecasts:

$$\widehat{\Delta y}_{t+1|t}^{\text{ens}} = \frac{1}{K} \sum_{k=1}^K \widehat{\Delta y}_{t+1|t}^{(k)}$$

where K is the count of non-naive models. The ensemble is then scored against naive and against the horse-race winner on the same metrics. This is the Bates and Granger (1969) idea, in its simplest form.

WHY AVERAGING WORKS IN MACRO FORECASTING

Stock and Watson (2004) build forecasts of output growth for seven countries with up to 73 predictors per country. Their main result:

- Individual forecasts are unstable over time and across countries; on average they underperform a simple autoregression.
- Combination forecasts often improve on autoregressive forecasts.
- Among combination methods, *the simplest combinations* — the equal-weight mean — are the most successful, because they are the *least sensitive* to recent in-sample performance.

Lesson: in macro forecasting, the simplest combination (“average the forecasts”) is often the best.

CASE STUDY 1: AUSTRALIA

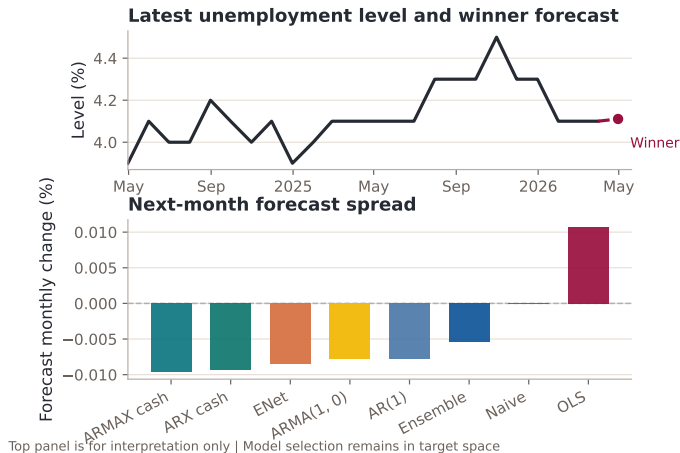
RECAP OF THE AUSTRALIA CASE

Same three rules, applied end-to-end:

- **Target:** monthly change in the Australia unemployment rate, one step ahead.
- **Split:** train through 2019-12-31, test from 2020-01-31.
- **Models:** naive, AR, ARMA, ARX, ARMAX, OLS, elastic net, horse race, equal-weight ensemble.
- **Data:** the Stage 1 long table from Week 2, ported from the `fins-agent` kit.

The script sequence in `fins2026/week3/scripts/` reproduces every leaderboard row and every figure on these slides; the 600-word interpretation lives in `fins2026/week3/README.md`.

AUSTRALIA: LATEST FORECAST



OLS dynamic regression under the default split.

- Solid line: realised monthly change.
- Marked point: the model's forecast for the next reference month.
- Inputs: target lags at $\{1, 3, 6, 12\}$ plus five exogenous predictors at $t - 1$.
- Fit on data through Dec 2019, rolled forward each month thereafter.
- Output also feeds the Streamlit app deployed in Section 14.

RESULT: OLS WINS

Under the 2019-12-31 / 2020-01-31 split, the OLS dynamic regression posts the best target-space RMSE and the best out-of-sample R^2 . Why this model, under this split?

- The five chosen exogenous predictors (cash rate, commodity, trade-weighted index, participation, vacancies) cover the main channels through which the Australian macroeconomy moves the unemployment rate.
- The COVID-and-after test window has large, fast moves — exactly the type of moves OLS picks up when the predictors carry signal.
- Adding the wider elastic-net bank does not help here: with extra predictors come extra correlations and the gain from shrinkage roughly offsets the loss from a bigger search.

Under a different split or with a different target, a different model would win, but the workflow does not change.

CASE STUDY 2: UNITED STATES

A SECOND CASE: U.S. UNEMPLOYMENT

We now mirror Case Study 1 on a second market: the United States.

- **Same target template:** monthly change in the U.S. unemployment rate, one step ahead.
- **Same split:** train through 2019-12-31, test from 2020-01-31.
- **Same evaluation contract:** target-space MAE, RMSE, MASE, out-of-sample R^2 versus naive.
- **Different data, different predictors:** FRED panel; the first ARX outside variable is the federal-funds change at $t - 1$, the second is the 10Y-2Y Treasury yield spread at $t - 1$.

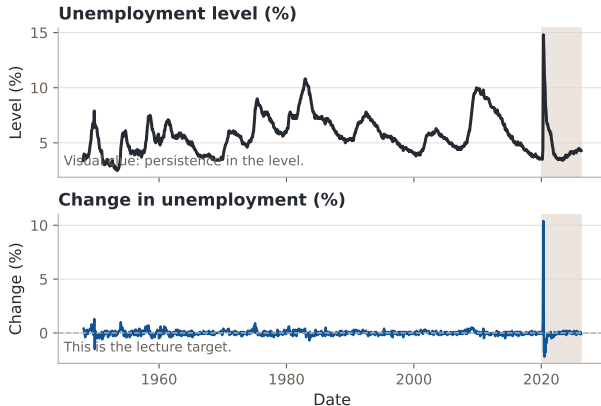
U.S. DATA SOURCES

The U.S. case uses the same no-key FRED graph CSV path introduced in Week 2. The lecture downloads:

- UNRATE — U.S. unemployment rate, monthly, seasonally adjusted
- FEDFUNDS — effective federal-funds rate, monthly average
- DGS10 and DGS2 — 10-year and 2-year Treasury yields, used to build the `yield_spread_pp = DGS10 - DGS2` series

Three real series, two transformed predictors. As in the Australia case, every exogenous predictor enters at $t - 1$.

U.S.: LEVEL VS CHANGE



Out-of-sample evaluation window shaded | Sample: 1948-01-31 to 2026-03-31 | Week 3 fixture data

Source: FRED UNRATE, monthly, seasonally adjusted.

- Top: level dominated by the COVID-19 spike to about 14% in April 2020.
- Outside the spike, the level is persistent — the unit-root signature.
- Bottom: first difference moves around zero with a stable variance, except for the April 2020 spike.
- ADF: cannot reject unit root in the level; reject in the change.
- Same diagnosis as Australia: model the change.

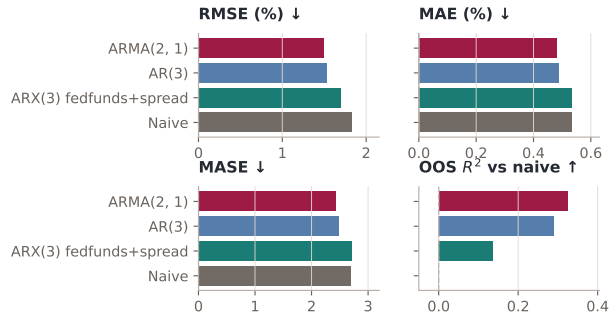
THE U.S. SET OF MODELS

The Australia case ran the full set of eight non-naive models because Australia is the primary case study. The U.S. case uses a shorter set by design:

- Naive
- $AR(p)$, BIC-selected
- $ARMA(p,q)$, BIC-selected
- $ARX(p)$ with two outside variables: fedfunds change, yield spread
- Horse race

Why shorter? The point of the U.S. case study is to show that the same workflow works on a new dataset — the three rules (change target, target-space evaluation, fixed split) apply equally to U.S. data. ARMAX, OLS, elastic net, and the ensemble are not needed to make that point.

Model evaluation scorecard

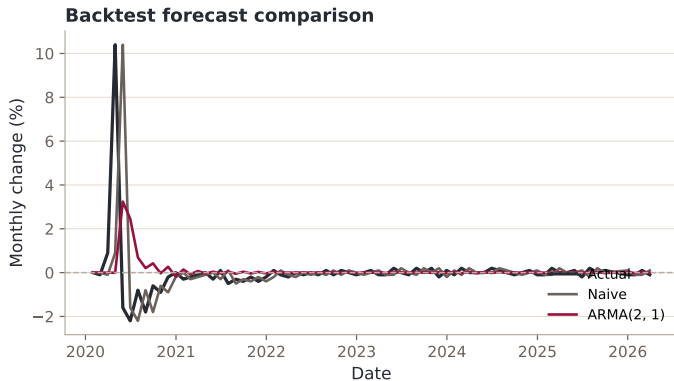


Lower is better for RMSE, MAE, and MASE. Higher OOS R^2 means a larger forecast-error reduction versus naive.

Four non-naive models, scored on Jan 2020 onward.

- **ARMA(2,1) wins:** target RMSE ≈ 1.50 pp; OOS $R^2 \approx 0.32$.
- AR(3) is a close second.
- ARX (fedfunds + 10Y-2Y spread) trails.
- Naive sits at the bottom.
- Different winner than Australia — workflow generalises, model does not.

U.S.: BACKTEST

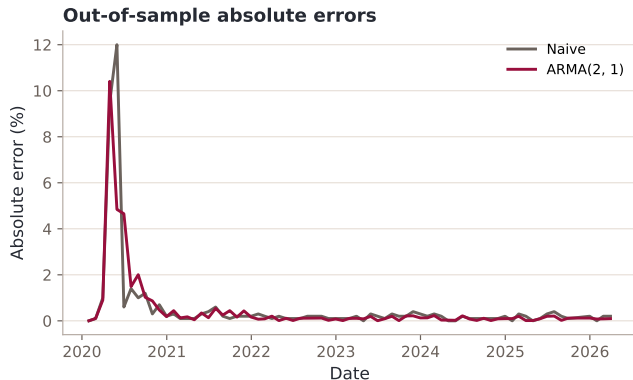


Out-of-sample only | Common evaluation dates: 73 months

Forecast paths vs realised U.S. unemployment change, Jan 2020 – Dec 2025.

- April 2020 spike of about four percentage points dominates the chart.
- ARMA(2,1) tracks the realised path most closely outside the spike.
- Naive (flat at zero) is visibly off through the COVID period.
- Ranking is stable across the rest of the test window.
- Same one-step-ahead rolling evaluation as the Australia case.

U.S.: ABSOLUTE ERRORS

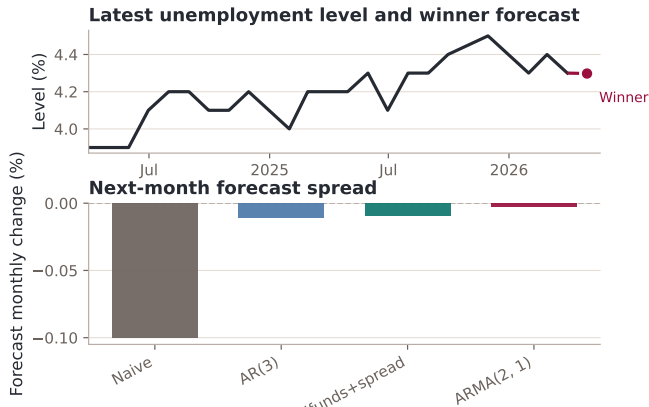


Lower lines mean smaller misses in the out-of-sample evaluation window

$|e_{t+1}|$ for each U.S. model, Jan 2020 onward.

- COVID spike dominates the error scale — a single hard month for every model.
- Ranking outside April 2020 mirrors the RMSE ranking on the leaderboard.
- ARMA(2,1) has the smallest typical miss, with AR(3) close behind.
- ARX errors widen in the post-COVID inflation episode.
- Naive remains the widest by visual inspection.

U.S.: LATEST FORECAST (WINNER: ARMA(2,1))



ARMA(2,1) on the FRED UNRATE change.

- Solid line: realised monthly change in U.S. unemployment.
- Marked point: ARMA(2,1) forecast for the next reference month.
- Inputs: two AR lags plus one MA term — no exogenous predictors.
- Fit on data through Dec 2019, rolled forward each month.
- U.S. companion app surfaces this number on its forecast tab.

Top panel is for interpretation only | Model selection remains in target space

THE ASSIGNMENT

REPEAT THE WORKFLOW ON A NEW TARGET

Your task: choose a macro variable from the **Australian or U.S. data we provide** (the same Stage 1 long table and FRED panel used in the two case studies). Your target must **not** be the unemployment rate.

Then repeat the workflow on the target you choose:

1. Define the target. Level, change, log change, or return?
2. Justify the transformation with an ADF check.
3. Set the default split: train through 2019-12-31, test from 2020-01-31.
4. Build the naive benchmark.
5. Fit AR, ARMA, and at least one ARX or OLS specification.
6. Report a horse-race leaderboard in target space.
7. Produce the same five FT-style narrative figures.
8. Write a 400-word interpretation.

SUGGESTED TARGETS

All eight targets below are available in the data we provide (Australia Stage 1 long table or the U.S. FRED panel). You can pick anything from the panels; these are good starting points.

- Headline CPI inflation, year-on-year (RBA G1; FRED CPIAUCSL)
- Real GDP growth, year-on-year (RBA H1; FRED GDPC1)
- 10-year Treasury yield, monthly change (RBA F2.1; FRED DGS10)
- Federal-funds rate, monthly change (FRED FEDFUNDS)
- Australian trade-weighted index, log change (RBA F11)
- Industrial production, log growth (FRED INDPR0)
- Wage Price Index growth, year-on-year (RBA H4)
- U.S. payroll employment, log change (FRED PAYEMS)

We grade process, not choice of target.

MARKING CRITERIA

- **Target:** did you justify the transformation with an ADF check, or did you forecast the level by reflex?
- **Split:** did you use the default 2019-12-31 / 2020-01-31 split, or did you justify a different one?
- **Benchmark:** is the naive benchmark scored, and is your winning model genuinely above it on the target?
- **Evaluation:** every metric in target space; no in-sample R^2 posing as the headline.
- **Figures:** the five FT-style figures, each captioned to stand alone (the four standards from Week 2).
- **Narrative:** 400 words, defending the transformation, the comparison, and the verdict.

We grade workflow, not the winning model. A careful execution of the model sequence on an ordinary target scores higher than a careless execution on an unusual one.

DEPLOY THE WEEK 3 APPS

WHAT YOU ARE DEPLOYING

Week 3 includes two Streamlit apps. Both are already built. Your job is to publish them cleanly, not to redesign them.

- **Australia macro forecast monitor**

`fins2026/week3/app/streamlit_app.py`

Wraps the leaderboard, the five Australia FT-style figures, the latest forecast, and the U.S. context tab.

- **U.S. macro stress and forecast monitor** (companion)

`fins2026/week3/us_app/streamlit_app.py`

The parallel U.S. surface — yield curve, GDP, forecast, backtest.

Run each app locally first, let your agent prepare the deploy repo, then publish to Streamlit Community Cloud.

RUN EACH APP LOCALLY FIRST

Before any deployment, confirm both apps run on your machine:

```
streamlit run fins2026/week3/app/streamlit_app.py
streamlit run fins2026/week3/us_app/streamlit_app.py
```

Each should open at `http://localhost:8501`. Step through the views and confirm the figures and tables match the case-study results in the lecture. Stop with `Ctrl+C` when you are done. A `localhost` address is for your machine only — it is not the public result.

Before deploying the Australia app, also run the pre-flight check:

```
python tools/workflow.py check-app-submission \
  --target fins2026/week3 \
  --entrypoint fins2026/week3/app/streamlit_app.py
```

LET YOUR AGENT PREPARE BOTH REPOS

Do not hand-build the deploy repository. Paste this prompt into your agent:

```
Use docs/apps/streamlit/student-quickstart.md and  
docs/apps/streamlit/finish-deployment.md to deploy  
both Week 3 Streamlit apps.
```

```
Keep both apps unchanged.
```

```
(1) Prepare and push week3-aus-forecast from  
fins2026/week3.  
Entrypoint: fins2026/week3/app/streamlit_app.py
```

```
(2) Prepare and push week3-us-macro from  
fins2026/week3.  
Entrypoint: fins2026/week3/us_app/streamlit_app.py
```

```
Then give me the exact Streamlit Cloud fields  
for each app and fix any deployment errors.
```

DEPLOY THE AUSTRALIA APP

On share.streamlit.io, create a new app from GitHub with:

- **Repository:** your-github-username/week3-aus-forecast
- **Branch:** main
- **Main file path:** fins2026/week3/app/streamlit_app.py
- **Python version:** 3.13

Leave secrets empty for Week 3.

- If the repo does not appear, reconnect GitHub under **Linked accounts** or use **Paste GitHub URL**
- After deploy, open **Settings** → **Sharing** and set **This app is public and searchable**
- Open the public `streamlit.app` URL in an incognito window to confirm

DEPLOY THE U.S. COMPANION APP

Create a second Streamlit Cloud app for the U.S. companion with:

- **Repository:** `your-github-username/week3-us-macro`
- **Branch:** `main`
- **Main file path:** `fins2026/week3/us_app/streamlit_app.py`
- **Python version:** `3.13`

Leave secrets empty.

- Both apps start in fixture mode and run offline. Switch to live mode in the sidebar when you want fresh RBA/FRED inputs.
- Do not commit `.streamlit/secrets.toml`. If a later-week app needs a key, add it in the Streamlit secrets UI, not the repository.
- Test both public URLs in an incognito window before submitting the assignment.

SUMMARY

RECAP: THE THREE RULES

Before you forecast anything, you must be able to say:

- **Target.** “I forecast the monthly change in X , one step ahead.”
- **Evaluation.** “I score every model on target MAE, RMSE, MASE, and out-of-sample R^2 versus the naive benchmark.”
- **Split.** “Train through 2019-12-31; test from 2020-01-31.”

These three sentences are the baseline of every forecasting deliverable in this course.

THE WORKFLOW SUMMARY

1. Choose a target and transformation. Justify with ADF.
2. Set the split.
3. Build the naive benchmark. Score it. This is the baseline every richer model must beat.
4. Work through the sequence of models: AR, ARMA, ARX, ARMAX, OLS, elastic net.
5. Each model is fit one-step-ahead on a rolling window. Each is scored in target space.
6. Report a horse-race leaderboard, ranked by target RMSE.
7. Add the equal-weight ensemble. Compare to the leaderboard winner.
8. Produce the five FT-style narrative figures.
9. Write the interpretation. Defend it.
10. Publish the working Streamlit app to Streamlit Community Cloud.

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